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First Group Assignment: STP Analytics: Ford Ka

In the early 1990s, car manufacturers segmented the car market using characteristics like vehicle size, engine output, and price. Cars were divided into categories mainly dependent on size where small cars were A/B, lower-medium was C, upper-medium was D, executive was E, and luxury was F. These product categories were associated with demographic customer segments. For example, small cars were generally targeted toward younger, lower-income consumers. Variables such as financial constraints and children were used to distinguish among small-car buyers. The traditional approach combined product-based segmentation with demographic segmentation based on age, income, and family.

As the small car market became increasingly fragmented, traditional B category cars began to be divided into 3 subcategories: the “Basic” B cars, the traditional small car, “Trend” B cars with better performance and more features, and “Other” B category cars, luxury small cars. The Fiesta was Ford’s entrant in the small car market. Later, in response to the small car market fragmentation, Ford released the “New” Fiesta, which was positioned as a comfortable, safe and dynamic car. Ford’s strategy was to make different versions of the Fiesta to target different parts of the market. The “New” Fiesta was targeted for Trend B consumers, but there was also a *Luxury* version and a *Family* version. The original Fiesta continued to be sold for the Basic B consumer. However, having multiple versions of the Fiesta wasn’t enough, as the market continued to fragment. The Renault Twingo gained a large amount of market share, despite originally targeting a narrow segment of buyers, due to its distinct style and greater functionality. Feeling challenged by the success of the Twingo and believing the Fiesta line lacked the appeal to compete, Ford decided to develop the Ka using the same chassis as the Fiesta. However, due

to time constraints, the key problem was that they had to find a target audience for an already designed product, rather than designing a product for an audience they knew existed.

Ford first conducted a two-stage market research study. First, it used focus groups with potential target customers to explore consumers' reactions and attitudes toward the Ford Ka, including first-time buyers, single working professionals, and multi-car households. Second, Ford conducted more structured personal interviews to understand how customers perceived the Ka relative to competing small cars. Respondents were asked to select their top three and bottom three choices from a list of ten cars, and they were also asked to use adjectives to describe each car. The research showed that the Ka did not clearly appeal to any specific demographic group. For example, younger consumers felt that the Ka was too much of an image risk, while older consumers often liked the car but thought it was designed for younger people. Ford also found that its traditional segmentation based mainly on age, income, and life stage did not clearly distinguish Ka Choosers from Non-Choosers. The company collected responses to 62 attitudinal questions to explore whether consumers could be grouped into psychographic segments. Overall, Ford wanted to collect information on customer preferences and broader attitudes to determine whether psychographic segmentation could identify a more appropriate target market than traditional demographic segmentation.

To facilitate comparison, age is converted from a continuous variable into six categories: under 25, 25–29, 30–34, 35–39, 40–44, and over 44. For each demographic variable, cross-tabulation is used to compare the number and percentage of consumers in each category between the two groups, Ka Choosers and Ka Non-Choosers. As shown in Table 1 (Appendix 1.1), the distributions of both the proportions and absolute numbers across the six demographic variables, Gender, Marital Status, First-Time Purchase, Age Category, Children Category, and

Income Category, show some differences between Ka Choosers and Ka Non-Choosers. However, these differences appear relatively limited and do not provide clear evidence that demographic characteristics can effectively distinguish between the two groups. Therefore, chi-square tests of independence were conducted to assess whether their associations with Ka preference are statistically significant. The significance level is set at **5% ($\alpha = 0.05$)**.

As shown in Table 2 (Appendix 1.1), none of the six demographic variables are significantly associated with preference for the Ford Ka at the **5%** significance level. All p-values are greater than **0.05**. Although Age Category has the largest chi-square statistic ($\chi^2 = 6.752$, $p = 0.24$), its p-value remains above the 0.05 threshold. Gender ($\chi^2 = 0.096$, $p = 0.76$) and First-Time Purchase ($\chi^2 = 0.000$, $p = 1.00$) show particularly little statistical evidence of an association with Ka preference. Therefore, the observed differences in demographic composition between Ka Choosers and Ka Non-Choosers are not statistically significant and there is insufficient evidence to conclude that demographic differences are associated with consumers' preference for the Ka.

We performed dimension reduction on the attitudinal variables by performing PCA to determine the number of factors. After performing PCA with no rotation and all components, we chose to keep 3 components. This choice was based on the elbow of the scree plot shown in Appendix 1.2 being 4. The variance is also greater than one, however it does not meet the criteria of having the cumulative variance be greater than 80%. In the end, it is still the best choice, since going above 3 components only has a small increase in cumulative variance and gives the most interpretable result. After performing PCA with rotation, we can analyze the 3 main factors we found and their loadings. The first factor is about style and self-expression with questions like

question 2 having high loadings and high communality. The second factor is about practicality and a dislike for small cars. The third factor revolves around a passion for cars and performance.

After identifying the factors, we performed K-means clustering on the factor scores to determine the number of clusters that the respondents fall into depending on the three factors found from PCA. We chose four segments based on Plot 2 in Appendix 1.2, since the elbow was at 4 and choosing any additional clusters would not give as distinct clusters. The resulting segments can be seen in Table 3. Cluster 0 is strongly positive for factor 3, therefore this cluster is composed of people who are hardcore car enthusiasts. Cluster 1 is positive for factor 2, with 65 of the respondents in this cluster. This segment cares about practicality and dislikes small cars. Cluster 2 has the majority of the respondents with 78, and is strongly positive for factor 1, indicating it is composed of those who care about style over practicality. Finally, the other 75 respondents fall under Cluster 3, which is negative for all three factors. These can be considered the average car drivers who don't have strong opinions on cars.

We recommend attitudinal segmentation, since demographic segmentation fails as there are no significant differences across demographics. In contrast, the attitudinal approach identifies 4 distinct clusters with strong statistical significance (**chi2 = 11.240, p = 0.0105 < 0.5**). Based on the cross-tabulation, we choose Cluster 0 and Cluster 2 as our primary target and secondary target, because they present a higher percentage of Ka chooser (**Cluster 0: 81.8%, 18 out of 22; Cluster 2: 72.3%, 34 out of 47**). Moreover, buyers in these clusters would likely resonate with Ka's innovative, avant-garde design and urban concept, so they are very likely high-potential target customers who have not yet been discovered by competitors like Renault and Peugeot.

In practice, psychographic data is not as readily available as demographic data; consequently, segmenting populations based on psychological data presents practical challenges.

Therefore, it would be difficult for Ford marketers to identify potential customers based on their psychographic profiles.

Appendix 1 Tables and Plots

Appendix 1.1 Tables for Question 4

Table 1: Demographic Composition of Ka Choosers and Ka Non-Choosers

Variable	Category	Chooser Count	Non-Chooser Count	Chooser %	Non-Chooser %
Gender	1	54	36	46.6	50.0
	2	62	36	53.4	50.0
Marital Status	1	66	34	56.9	47.2
	2	14	6	12.1	8.3
	3	36	32	31.0	44.4
1st Time Purchase	1	13	8	11.2	11.1
	2	103	64	88.8	88.9
Age Category	1	10	3	8.6	4.2
	2	18	13	15.5	18.1
	3	23	12	19.8	16.7
	4	11	11	9.5	15.3
	5	36	15	31.0	20.8
	6	18	18	15.5	25.0
Children Category	0	62	45	53.4	62.5
	1	29	12	25.0	16.7
	2	25	15	21.6	20.8
Income Category	1	11	5	9.5	6.9
	2	19	15	16.4	20.8

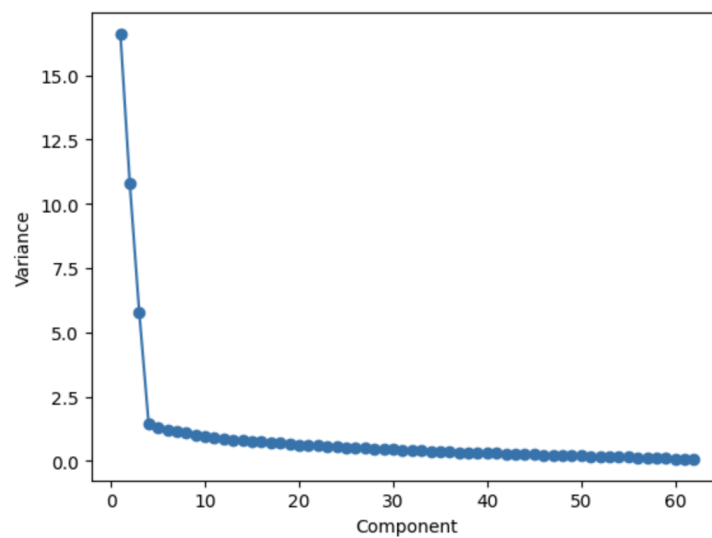
3	18	16	15.5	22.2
4	19	16	16.4	22.2
5	28	12	24.1	16.7
6	21	8	18.1	11.1

Table 2: Chi-Square Tests of Independence between Demographic Characteristics and Ford Ka Preference

Variable	Chi2	p	dof
Gender	0.096	0.76	1
Marital Status	3.573	0.17	2
1st Time Purchase	0	1.00	1
Age Category	6.752	0.24	5
Children Category	2.065	0.36	2
Income Category	5.316	0.38	5

Appendix 1.2: Tables and Plots for Question 5

Plot 1: Scree plot for PCA without Rotation



Plot 2: Elbow Plot for Clustering

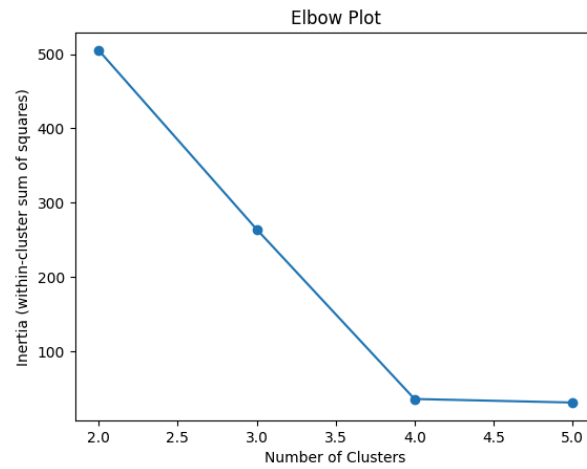


Table 3: K-means Analysis

	RC1	RC2	RC3
0	-0.464984	-0.533837	2.405634
1	-0.200742	1.643329	-0.056442
2	1.383266	-0.430916	-0.245159
3	-1.066228	-0.748296	-0.722523

Appendix 1.3: Tables for Question 6

Table 4: Chi-square Test Results between Psychographic Clusters and Ka-Choosers

Cluster vs Preference Group				
Cluster	1 Count	2 Count	1 %	2 %
0	18	4	81.8	18.2
1	29	32	47.5	52.5
2	34	13	72.3	27.7
3	35	23	60.3	39.7

Cluster vs Preference Group				
Chi2 = 11.240				
p = 0.0105				
dof = 3				